

UEC633: NETWORK SECURITY AND CRYPTOGRAPHY

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2	0	2	3.0

Course Objective: This course is designed to impart a critical theoretical and detailed practical knowledge of cryptographic algorithms and techniques. To recognize typical vulnerabilities and safeguards used for wireless communication.

Basic of Cryptography: Symmetric and asymmetric cryptography, cryptographic hash functions, Authenticated encryption, Key generation and distribution, Key management, Message Authentication Codes (MACs), Message integrity, Message Authentication and Hash Function: Authentication requirements.

Symmetric cryptography: Introduction to DES, TDES and AES algorithms, Blowfish and Twofish. Public key cryptography: Introduction to Diffie-Hellman, RSA, ECC, ElGamal and DSA, Lightweight encryption.

Digital signatures: How to sign using RSA, Hash based signatures, certificates, certificate transparency, certificate revocation.

Security Vulnerabilities: security against active attacks, Attacks under Message Indistinguishability (Chosen Plaintext Attack and Chosen Ciphertext Attacks), middle channel and side channel attacks, Access Control Problems, Spoofing and Sniffing attacks, Phishing attacks, Ransomware, zero-day attacks, birthday attacks, password attacks, insider attacks, Malware and Social Engineering and countermeasures.

Wireless Network Security: IEEE 802.11i Wireless LAN Security, Wireless Application Protocol Overview, Wireless Transport Layer Security, WAP End-to-End Security.

Familiarization with standards: IEEE 1363-2000, IEEE Standard Specifications for Public-Key Cryptography, IEEE 1363.2-2008, IEEE Standard Specification for Password-Based Public Key Cryptographic Techniques, IEEE 1363.3-2013, IEEE Standard for Identity-Based Cryptographic Techniques using Pairings

Laboratory Work: Write program to send data across systems as two variants clear text data and encrypted data with different set of encryption algorithms, Showcase different set of security protocol implementation of Wireless LAN.

Course Learning Outcomes (CLO): The student will be able to

1. Implement various cryptographic algorithms to protect the confidential data.
2. Identify network vulnerabilities and apply various security mechanisms to protect networks from security attacks.
3. Apply security tools to locate and fix security leaks in a computer network/software.
4. Secure a web server and web application.

Text Books:

1. Stallings, W., Network Security Essentials, Prentice Hall (2017) 6th Edition.
2. Cheswick, R., W., Bellovin, M., S., and Rubin, D., A., Firewalls and Internet Security, Addison-Wesley Professional (2003) 2nd Edition.

Reference Books:

1. Graves, K., Certified Ethical Hacking Study Guide, Sybex (2010) 1st Edition.
2. Cryptography and Network Security Principles and Practice (2020), 8th Edition – Pearson